The Informational Efficiency of Frequency-Report Scoring Rules

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Abstract

In a multinomial frequency-report environment, a subject has latent beliefs p, but the researcher observes only an incentivized count report r. We compare frequency-report scoring rules by how sharply that report identifies the latent beliefs. The inference is recast as a partial-identification problem: each scoring rule maps the report to an identified set of beliefs—those under which the report would have been optimal—from which coordinate and mean bounds follow. The main analytical result is that quadratic-distance frequency scoring gives a transparent projection rule, a linear identified set, closed-form coordinate bounds, and linear-program bounds for means. Discrete-metric scoring is the known fixed-prize rule with closed-form bounds, and Manhattan distance has an exact identified set with sharp semi-analytical bounds. A design comparison over these three rules finds no universal ranking: the most informative rule depends on the inferential objective and the design regime.

1 Introduction

Belief elicitation in experiments often asks subjects to report probabilities directly. Frequency reports ask instead for counts: out of n realizations, how many times will each outcome occur? This format is natural when the object of interest is a distribution of choices, signals, or outcomes in a finite group. It may also be easier for subjects than reporting abstract probabilities under a standard scoring rule. Many probability scoring rules, including the quadratic scoring rule, are well understood theoretically (Savage, 1971; Gneiting and Raftery, 2007; Selten, 1998), but they require subjects to understand how probability reports map into payments. That mapping can be cognitively demanding, especially when subjects reason more naturally in frequencies than in abstract probabilities (Hogarth, 1975; Schlag et al., 2015). Direct monetary scoring rules also raise the familiar risk-aversion problem (Armantier and Treich, 2013).

The practical problem is that the researcher observes a count report r, not the subject's latent belief vector p. The report is an incentivized message generated by a scoring rule. Thus the inferential question is inverse: given the rule and the observed report, which beliefs are consistent with optimal reporting? The paper follows the chain

$$S \;\to\; R_S(p) \;\to\; P_S(r) \;\to\; \left[\underline{p}_{S,i}(r), \overline{p}_{S,i}(r)\right].$$

Here S is a frequency-report scoring rule, $R_S(p)$ is the set of reports that are optimal for a subject with beliefs p, and

$$P_S(r) = \{p \in \Delta^k : r \in R_S(p)\}$$

is the identified set induced by the observed report. This recasts incentivized belief elicitation as a partial-identification problem (Manski, 2003): the scoring-rule mechanism, rather than a sampling distribution over repeated observations, supplies the identifying restriction. The final object of the chain is the bound $[\underline{p}_{S,i}(r), \overline{p}_{S,i}(r)]$ it yields on each individual probability p_i. A researcher may also care about functionals of the belief vector, such as a mean or another linear index; bounds on these are obtained from the same identified set $P_S(r)$.

This paper uses the term informational efficiency in an operational sense. A rule is more informative for a given design objective if the observed report leads to narrower coordinate bounds or narrower mean and functional bounds. These inferential gains must be weighed against implementation costs: exact-match payment probabilities, cognitive simplicity, computational burden, and robustness to risk aversion.

The paper's main analytical contribution is for quadratic-distance frequency scoring. Under this rule, a subject reports the feasible count vector closest to the expected count vector np. This gives a linear identified set and closed-form bounds for each p_i. Mean bounds and other linear functional bounds are then simple linear programs over the same region.

The paper compares this rule to two other count-loss mechanisms along a spectrum of analytical tractability. Schlag–Tremewan frequency guessing is the known discrete-metric exact-match mechanism: it elicits multinomial modes, gives closed-form bounds, and is robust to risk aversion under a fixed-prize implementation (Schlag and Tremewan, 2021). Manhattan-distance scoring has an exact identified set and sharp, semi-analytical coordinate bounds.

The contribution is not a new exact-match mechanism and not a general theory of scoring rules; Schlag and Tremewan already derive belief bounds for the frequency-guessing rule. It is, first, the characterizations themselves: a closed-form identified set and coordinate bounds for quadratic-distance scoring, the centerpiece result, and sharp semi-analytical bounds for Manhattan distance, alongside the known discrete-metric bounds. It is, second, the contingent decision rule those characterizations make rigorous: in a design comparison over the three rules with tractable bounds, the most informative rule depends on the inferential objective and the design regime, with no universal ranking. Partial identification is the lens that holds these together—each scoring rule maps an observed report to an identified set of latent beliefs—and is what lets the comparison be posed as a sharp, finite-sample question rather than an informal one. Two further count-loss rules, Hamming and Chebyshev distance, mark the boundary of the approach; Section 5 discusses them as rules whose identified sets are exact but resist tractable sharp bounds.

2 Setup

The environment is a finite belief-elicitation problem. There are k possible outcomes, and the subject has latent beliefs

$$p = (p_1, \ldots, p_k) \in \Delta^k, \quad p_i \geq 0, \quad \sum_{i=1}^{k} p_i = 1.$$

The elicitation task asks about $n \in \mathbb{N}$ independent repetitions. If the subject's beliefs are p, the realized count vector is

$$\omega = (\omega_1, \ldots, \omega_k) \sim \text{Mult}(n, p).$$

The set of feasible count vectors is

$$\Omega = \left\{ \omega \in \mathbb{N}_0^k : \sum_{i=1}^{k} \omega_i = n \right\},$$

and the multinomial probability mass function is

$$\Pr_p(\omega) = \frac{n!}{\omega_1! \cdots \omega_k!} \prod_{i=1}^{k} p_i^{\omega_i}.$$

The subject reports another feasible count vector

$$r = (r_1, \ldots, r_k) \in \Omega.$$

Definition 1 (Frequency-report scoring rule)**.** A frequency-report scoring rule is a function

$$S : \Omega \times \Omega \to \mathbb{R},$$

where $S(r, \omega)$ is the score assigned to report r when the realized count vector is ω. The paper studies rules that reward reports close to realized counts. A convenient form is

$$S_D(r, \omega) = a - bD(r, \omega), \quad b > 0,$$

where $D : \Omega \times \Omega \to \mathbb{R}_+$ is a count-error index. The function D need not be a metric: squared distance is a divergence-style count loss, not a metric, because it fails the triangle inequality.

The constants a and b do not matter for expected-score rankings as long as $b > 0$. They matter for implementation. If scores are paid directly as money and nonnegative payments are desired, one can choose

$$a \geq b \max_{r, \omega \in \Omega} D(r, \omega).$$

How the score is implemented as a payment affects robustness to the subject's risk attitude; the paper's discussion of risk aversion and implementation takes up this question separately.

Definition 2 (Optimal reports)**.** Under risk neutrality, the optimal-report correspondence induced by a scoring rule S is

$$R_S(p) = \arg\max_{r \in \Omega} \mathbb{E}_p[S(r, \omega)].$$

Because Ω is finite, $R_S(p)$ is nonempty.

Risk neutrality is the maintained assumption throughout the body of the paper; the discussion of risk aversion and implementation separately shows how the analysis extends to risk-averse expected-utility subjects.

This is the forward incentive object. It answers the question: if the subject's latent beliefs are p, which reports are optimal? The econometric object is the inverse question. After observing r, the researcher asks which beliefs could have generated that report.

Definition 3 (Identified set)**.** For an observed report $r \in \Omega$, the identified set induced by S is

$$P_S(r) = \{p \in \Delta^k : r \in R_S(p)\}.$$

Ties are included: if r is one of several optimal reports under p, then $p \in P_S(r)$.

We call $P_S(r)$ the *mechanism-induced identified set*: the identifying restriction is the optimal-report condition $r \in R_S(p)$ imposed by the scoring-rule mechanism, not a sampling distribution over repeated observations. By construction $P_S(r)$ is exactly the set of beliefs consistent with that restriction.

The most direct summaries of $P_S(r)$ are coordinate-wise identification bounds:

$$\underline{p}_{S,i}(r) = \inf_{p \in P_S(r)} p_i, \quad \overline{p}_{S,i}(r) = \sup_{p \in P_S(r)} p_i,$$

with interval

$$P_{S,i}(r) = [\underline{p}_{S,i}(r), \overline{p}_{S,i}(r)].$$

When the scoring rule is clear from context, the subscript S is omitted.

Many experiments require more than coordinate-by-coordinate probability bounds. If outcomes have numerical values $x_1, \ldots, x_k$, the latent mean implied by beliefs p is

$$\mu(p; x) = \sum_{i=1}^{k} p_i x_i.$$

This could represent an expected outcome, an expected treatment share, a predicted group average, or another linear index of the belief distribution. The sharp mean bounds after observing r are

$$\underline{\mu}_S(r; x) = \inf_{p \in P_S(r)} \sum_{i=1}^{k} p_i x_i, \quad \overline{\mu}_S(r; x) = \sup_{p \in P_S(r)} \sum_{i=1}^{k} p_i x_i.$$

The corresponding mean interval is

$$[\underline{\mu}_S(r; x), \overline{\mu}_S(r; x)].$$

These bounds generally require optimizing over the full joint region $P_S(r)$. Coordinate intervals alone do not generally give sharp bounds for a mean, because they ignore dependence across coordinates imposed by the simplex and the scoring rule.

The design comparison later summarizes the informativeness of a rule using coordinate widths,

$$\overline{p}_{S,i}(r) - \underline{p}_{S,i}(r),$$

and functional widths,

$$\overline{\mu}_S(r; x) - \underline{\mu}_S(r; x).$$

Exact-match payment probability is reported only for the discrete-metric rule, as an implementation diagnostic of that fixed-prize mechanism; it is not a cross-rule comparison metric.

The next section applies this setup to the three headline rules: quadratic-distance, discrete-metric, and Manhattan.

3 Frequency-Report Scoring Rules

This section applies the setup to three frequency-report scoring rules. They span a spectrum of analytical tractability. Quadratic-distance and discrete-metric scoring have closed-form identified sets and closed-form coordinate bounds. Manhattan-distance scoring has an exact identified set and sharp coordinate bounds in semi-analytical form. The same machinery applies to further count-loss criteria—Hamming and Chebyshev (L_∞) distance—whose identified sets are exact but resist tractable sharp bounds; the discussion takes them up as the boundary of the approach.

3.1 Rules with Closed-Form Bounds

3.1.1 Quadratic-Distance Frequency Scoring

The main analytical rule is quadratic-distance frequency scoring. It pays less when the reported count vector is far from the realized count vector in squared Euclidean distance:

$$S_Q(r,\omega) = a - b\|r-\omega\|_2^2, \quad b>0.$$

This rule is connected in spirit to quadratic scoring rules, but it is not a standard probability score. It is a frequency-report count-loss rule: subjects report counts, and the penalty is the squared difference between reported and realized counts. Write P_Q for the identified set induced by this rule.

Given beliefs p, the expected count vector is np. The key forward fact is that quadratic-distance scoring asks the subject to report the feasible integer count vector closest to np. The inverse question is therefore geometric: for which beliefs is the observed report r the closest feasible count vector to np?

Proposition 1 (Quadratic distance: reports, regions, and bounds)**.** *For every $p \in \Delta^k$,*

$$R_{S_Q}(p) = \arg\min_{s\in\Omega} \|s-np\|_2^2.$$

For an observed report $r \in \Omega$, the identified set is

$$P_Q(r) = \left\{ p \in \Delta^k : n(p_i - p_j) \le r_i - r_j + 1 \quad \forall i,j \text{ with } r_j > 0 \right\}.$$

Let

$$m(r) = |\{j : r_j > 0\}|$$

be the number of categories with positive reported counts. The sharp coordinate intervals are

$$P_{Q,i}(r) = \begin{cases} \left[0, \dfrac{m(r)}{n(m(r)+1)}\right], & r_i = 0, \\[2ex] \left[\dfrac{r_i-1}{n} + \dfrac{1}{nk}, \dfrac{r_i+1}{n} - \dfrac{1}{n\,m(r)}\right], & r_i > 0. \end{cases}$$

Projection ties are included.

The inequalities compare the observed report r with reports obtained by moving one count from a positive reported category j to another category i. If all such one-count moves weakly increase squared distance from np, then r is a closest feasible report. Thus $P_Q(r)$ is a polytope in the belief simplex. The proof is in Appendix A.3.

Probability and mean bounds. The coordinate bounds are closed form. For numerical outcomes $x_1, \ldots, x_k$, sharp mean bounds are obtained by optimizing over the same polytope:

$$\min_{p \in P_Q(r)} \sum_{i=1}^{k} p_i x_i \quad \text{and} \quad \max_{p \in P_Q(r)} \sum_{i=1}^{k} p_i x_i.$$

These are linear programs. For coordinate probabilities, the displayed intervals give the corresponding linear-program solutions explicitly. This is the main practical advantage of quadratic-distance frequency scoring: the same transparent region gives both probability bounds and bounds on expected values.

Practical interpretation. Quadratic distance is mean-oriented: it links a report r to beliefs for which r is the integer projection of np. Compared with exact matching, the payment varies smoothly with distance rather than only with exact equality. The main implementation cost is that direct monetary quadratic-distance payments generally require risk neutrality unless the score is implemented probabilistically, as discussed below.

3.1.2 Discrete-Metric Frequency Scoring

The second closed-form rule is the frequency-guessing mechanism of Schlag and Tremewan (2021). Its count-error index is the count-vector discrete metric,

$$D_0(r, \omega) = \mathbf{1}\{r \neq \omega\},$$

so, in the form $S_D = a - bD$ of Section 2, the score is

$$S_0(r, \omega) = a - b\,\mathbf{1}\{r \neq \omega\}, \quad b > 0.$$

Equivalently, $S_0(r, \omega) = (a - b) + b\,\mathbf{1}\{r = \omega\}$: the subject receives an extra b whenever the realized count vector equals the report. This fixed-prize reading is useful for implementation and risk aversion; the two formulations induce the same optimal reports. Write R_0 and P_0 for the report correspondence and identified set induced by this rule.

The mechanism is simple: a subject with beliefs p maximizes the probability of an exact match,

$$\Pr_p(\omega = r) = \frac{n!}{r_1! \cdots r_k!} \prod_{i=1}^{k} p_i^{r_i}.$$

Thus the forward object is the multinomial mode set. The inverse object is the set of beliefs for which the observed report is one of those modes.

Proposition 2 (Discrete metric: reports, regions, and bounds)**.** *Let $r \in \Omega$, and allow ties in the multinomial mode. Then r is optimal under discrete-metric frequency scoring if and only if r is a multinomial mode:*

$$R_0(p) = \arg\max_{s \in \Omega} \Pr_p(\omega = s).$$

The identified set is

$$P_0(r) = \left\{ p \in \Delta^k : r_j p_i \leq (r_i + 1) p_j \quad \forall i, j \text{ with } r_j > 0 \right\}.$$

For each coordinate, the sharp identification interval is

$$P_{0,i}(r) = \left[\frac{r_i}{n+k-1}, \frac{r_i+1}{n+1}\right].$$

The inequalities say that no one-count transfer from a reported category j to another category i can increase the multinomial probability of the report. Weak inequalities include ties. The formula also handles boundaries: if $r_i = 0$, the lower bound is 0; if $r_i = n$, the upper bound is 1. These are the frequency-guessing bounds in Schlag and Tremewan (2021, Proposition 1); they are restated here to make the informational-efficiency comparison self-contained. The proof is in Appendix A.1.

Probability and mean bounds. The discrete-metric rule gives closed-form coordinate intervals. For linear functionals such as means, sharp bounds use the full joint region, not just the coordinate intervals:

$$\min_{p\in P_0(r)} \sum_{i=1}^{k} p_i x_i \quad \text{and} \quad \max_{p\in P_0(r)} \sum_{i=1}^{k} p_i x_i.$$

These are linear programs because $P_0(r)$ is defined by linear inequalities. Thus discrete-metric frequency scoring is analytically convenient for individual probability bounds, while mean inference still requires using the joint mode region.

Practical interpretation. Discrete-metric frequency scoring is transparent and robust to risk aversion because it is a fixed-prize event. Its implementation cost is that the probability of an exact match can be small when n is large or the outcome space is rich. It is a prominent known rule in the comparison, not a new mechanism introduced here.

3.1.3 Comparing the Two Closed-Form Rules

The two closed-form rules can be compared directly through their coordinate-bound widths. The discrete-metric coordinate width, $[r_i(k-2) + (n+k-1)]/[(n+1)(n+k-1)]$, is linear in the reported count r_i; because $\sum_i r_i = n$ fixes the average reported count at n/k, the average of this width over the k coordinates is a constant,

$$\frac{(k-1)(2n+k)}{k(n+1)(n+k-1)},$$

the same for every report. The quadratic-distance coordinate width for a positive-report category, $2/n - 1/(nm) - 1/(nk)$ with $m = m(r)$ the number of categories with positive reported counts, does not depend on r_i and shrinks as the report concentrates on fewer categories. A report concentrated on few categories therefore has a small average quadratic width; a balanced report, with $m = k$, has an average quadratic width that exceeds the discrete-metric constant. Which of the two closed-form rules gives the narrower average coordinate bound thus turns on how concentrated the report is, before any simulation—a contrast the design comparison of the next section ties to belief concentration, quantifies, and extends to Manhattan distance.

3.2 Manhattan-Distance Frequency Scoring

Unlike the two closed-form rules, Manhattan-distance scoring has no closed-form coordinate bounds; its identified set is nonetheless exact, and its sharp bounds reduce to a one-dimensional computation. The Manhattan-distance rule is

$$S_M(r, \omega) = a - b \sum_{i=1}^{k} |r_i - \omega_i|, \quad b > 0.$$

It rewards total absolute count accuracy rather than squared count accuracy. Its inferential interpretation is median-oriented rather than mean-oriented.

Optimal reports and identified set. For category i, the realized count has binomial marginal distribution

$$W_i \sim \mathrm{Bin}(n, p_i).$$

Let

$$\ell_i(t; p_i) = \mathbb{E}|t - W_i|, \qquad F_i(t; n, p_i) = \Pr(W_i \le t).$$

Then optimal reports solve

$$R_M(p) = \arg\min_{r \in \Omega} \sum_{i=1}^{k} \ell_i(r_i; p_i).$$

The simplex constraint $\sum_i r_i = n$ prevents the researcher from treating each coordinate independently. Because each coordinate loss ℓ_i is discrete-convex and the objective is separable, the absence of any single-count transfer that lowers expected Manhattan loss is both necessary and sufficient for global optimality. Equivalently,

$$P_M(r) = \left\{ p \in \Delta^k : F_i(r_i; n, p_i) \ge F_j(r_j - 1; n, p_j) \quad \forall i, j \text{ with } r_i < n,\ r_j > 0 \right\}.$$

Weak inequalities include ties; there is no receiver constraint when $r_i = n$ and no sender constraint when $r_j = 0$. Single-unit-transfer optimality is sufficient, not merely necessary, so $P_M(r)$ is the exact identified set rather than an outer approximation. The proof is in Appendix A.2.

Identification bounds. For $k = 2$, write $r = (t, n - t)$. Then Manhattan loss is $2|t - W|$, with $W \sim \mathrm{Bin}(n, p_1)$, so the report identifies the binomial-median interval

$$P_{M,1}(r) = \{p_1 \in [0, 1] : F(t - 1; n, p_1) \le 1/2 \le F(t; n, p_1)\}.$$

This binary case is clean.

For $k > 2$, the pairwise CDF inequalities are equivalent to a single threshold condition: $p \in P_M(r)$ if and only if there exists $c \in [0, 1]$ such that

$$F_i(r_i - 1; n, p_i) \le c \le F_i(r_i; n, p_i) \quad \forall i,$$

using $F_i(-1; n, p_i) = 0$ and $F_i(n; n, p_i) = 1$. For a fixed c, these inequalities give a coordinate box through inverse binomial CDFs, and intersecting that box with the simplex gives the feasible beliefs at that threshold. Each sharp coordinate bound then reduces to a one-dimensional optimization over the threshold c: the objective is unimodal in c, so the bound is the solution of a single monotone scalar equation. The coordinate bounds are thus sharp and semi-analytical—not closed form, because the inverse binomial CDF is not elementary, but exact and one-dimensional rather than grid-limited. The sharp mean bounds are obtained by a one-dimensional search over the same threshold.

Practical interpretation. Manhattan distance is useful when a median-type count prediction is substantively appropriate. It is also useful as a contrast: changing the distance criterion changes the inferred identified set. The multi-category Manhattan bounds are sharp and reduce to a one-dimensional threshold computation; they are less transparent than the closed-form quadratic-distance bounds but are exact, not approximations.

4 Informational Efficiency: A Design Comparison

The analytical regions are useful because they tell an experimenter what can be inferred after observing a report. This section reports a design comparison generated by the reproducible script `scripts/design_efficiency.py`. The exercise draws latent beliefs

$$p \sim \text{Dirichlet}(\alpha, \dots, \alpha)$$

and, for each of the three rules with tractable sharp bounds—discrete-metric, quadratic-distance, and Manhattan distance—computes an optimal report $r_S(p)$ and the bounds induced by $P_S(r_S(p))$. Hamming and Chebyshev distance fall outside this comparison; Section 5 takes them up as count-loss rules whose identified sets are exact but resist tractable sharp bounds. The final run uses 5,000 belief draws for each cell in

$$n \in \{5, 10, 20, 50\}, \qquad k \in \{2, 3, 5, 10\}, \qquad \alpha \in \{0.1, 0.3, 1, 3, 10\}.$$

The Dirichlet parameter α sets belief concentration: small α draws sparse beliefs near a vertex of the simplex, large α draws balanced, interior beliefs. For each rule the comparison records two headline inferential targets—the average width of the coordinate bounds, and the width of the bound on the linear functional $\sum_i p_i x_i$ with outcome values $x_i = (i-1)/(k-1)$. Manhattan coordinate bounds are sharp, computed by a threshold root-find; Manhattan mean bounds are threshold-computed with tolerance 10^{-4}.

The comparison is conditional on the theoretical characterizations above. It is not evidence of incentive compatibility, and it is not an optimality theorem. It plays three roles: it confirms the analytic crossover between the two closed-form rules established in Section 3, quantifies that crossover through win shares and regret, and extends the comparison to Manhattan distance, whose lack of a closed-form bound rules out a comparable analytic treatment. It is, in short, a diagnostic for the practical question: which rule gives a researcher tighter inference for the object of interest?

The ranking is governed by belief concentration. Section 3 showed, from the closed-form bounds alone, that which of the two closed-form rules gives the narrower average coordinate bound turns on report concentration—the number m of categories with positive reported counts. The design comparison shows that report concentration tracks belief concentration, so the crossover is governed by how concentrated the subject's beliefs are. Averaged over the whole grid the three rules barely differ—mean coordinate width 0.103, 0.104, and 0.102 for discrete-metric, quadratic-distance, and Manhattan scoring—because each rule's favourable and unfavourable regimes cancel. Table 1 reports the win shares—the share of belief draws on which a rule gives the narrowest interval—broken out by α.

	Average-coordinate			Linear-mean		
α	Discrete	Quadratic	Manhattan	Discrete	Quadratic	Manhattan
0.1	0.08	0.84	0.07	0.08	0.83	0.10
0.3	0.28	0.58	0.14	0.22	0.60	0.18
1	0.76	0.16	0.08	0.58	0.27	0.15
3	0.98	0.01	0.01	0.83	0.13	0.04
10	1.00	0.00	0.00	0.92	0.06	0.02

Table 1: Win share by belief concentration α: the share of belief draws on which a rule gives the narrowest interval, averaged over all n and k at each α. Quadratic-distance scoring wins for sparse beliefs (small α); discrete-metric scoring for balanced beliefs (large α).

Quadratic-distance scoring is the most informative rule when beliefs are sparse and discrete-metric scoring when they are balanced, with a crossover near $\alpha = 1$. At $\alpha = 0.1$ quadratic gives the narrowest average-coordinate interval on 0.84 of draws; by $\alpha \geq 3$ discrete-metric does so on essentially every draw. The linear-mean target shows the same crossover. The number of categories k and the number of trials n are second-order, as Table 3 in Appendix B documents: for the coordinate target the ranking is nearly flat in both, while for the mean target larger n mildly favours quadratic and the binary case $k = 2$ favours discrete-metric. The worst-coordinate target follows the same pattern shifted in quadratic's favour—quadratic gives the narrowest maximum-coordinate interval on a majority of draws for every $\alpha \leq 3$. The ranking is robust to the choice of outcome vector and is unaffected by optimal-report ties, which are negligible under the Dirichlet draws; Appendix B reports both checks.

The crossover has a mechanism in the projection interpretation of quadratic-distance scoring. A subject reports the integer count vector closest to np; when beliefs are sparse, np lies near a vertex of the simplex, the feasible reports around it are extreme count vectors, and the projection polytope is tight, so the induced bounds are narrow. As beliefs become balanced, np moves into the interior, the projection polytope widens, and discrete-metric scoring—which elicits the multinomial mode—becomes the more informative rule.

The cost of the wrong rule, and Manhattan as a hedge. Win shares count only first places. Cell-best regret—a rule's interval width minus the narrowest width available in the same cell—measures how much precision is forfeited by not using the best rule. Table 2 reports regret by α.

Discrete-metric and quadratic-distance scoring each have a regime of near-zero regret and a regime in which regret is substantial. Discrete-metric's average-coordinate regret runs from 0.000 at $\alpha = 10$ to 0.032 at $\alpha = 0.1$; quadratic's runs the other way, reaching 0.024 on the mean target at $\alpha = 10$. Against typical interval widths near 0.10 to 0.15, the worst-regime values are a quarter or more of an interval width—a real loss of precision from choosing the rule that does not match the belief regime. Manhattan distance behaves differently. It is never the narrowest rule, but its regret is nearly constant in α—between 0.007 and 0.016 at every

	Average-coordinate			Linear-mean		
α	Discrete	Quadratic	Manhattan	Discrete	Quadratic	Manhattan
0.1	0.032	0.002	0.011	0.047	0.002	0.013
0.3	0.016	0.005	0.007	0.024	0.005	0.009
1	0.003	0.012	0.007	0.006	0.013	0.008
3	0.000	0.019	0.011	0.001	0.020	0.012
10	0.000	0.023	0.014	0.000	0.024	0.016

Table 2: Mean cell-best regret by belief concentration α: a rule's interval width minus the smallest width among the three rules, averaged over all n and k at each α. Lower is better. Discrete-metric and quadratic scoring each swing between near-zero and substantial regret; Manhattan's regret is nearly flat.

concentration and for both targets—so it is never far from the best rule either. Manhattan is therefore the regime-robust choice: a researcher who cannot anticipate how concentrated subjects' beliefs are minimises worst-case regret by using it, whereas committing in advance to discrete-metric or quadratic scoring risks the unfavourable regime. This is the precise sense in which Manhattan is a low-regret rule: it hedges the belief-concentration uncertainty to which the other two rules are exposed.

Payment probability. The discrete-metric rule has a fixed-prize implementation, which is attractive under risk aversion. Its implementation cost is that the exact-match payment probability can be small. In the final run, the average exact-match probability is high for very sparse binary beliefs, but it falls sharply as n, k, and belief balance increase; for instance, it is essentially zero in the $n = 50, k = 10, \alpha \in \{1, 3, 10\}$ cells. Thus narrow bounds under discrete-metric scoring should be weighed against the chance that the subject rarely wins the fixed prize.

The comparison thus delivers a contingent recommendation, not a single winner: the most informative rule is the one matched to the anticipated belief regime and inferential target, a choice the discussion turns into practical guidance.

5 Discussion

The paper studies frequency-report scoring rules as inferential devices. The report r is not the belief vector. It is evidence about p, and the scoring rule determines how informative that evidence is. The relevant chain is therefore

$$S \;\rightarrow\; R_S(p) \;\rightarrow\; P_S(r) \;\rightarrow\; \big[\underline{p}_{S,i}(r), \overline{p}_{S,i}(r)\big].$$

The design comparison shows that the most informative rule depends on how concentrated the subject's beliefs are; three practical recommendations follow. First, use discrete-metric frequency scoring when the subject is expected to spread probability fairly evenly across the outcomes, with no single outcome carrying most of the probability. It is the most informative rule in that regime, has closed-form coordinate bounds, and is the only rule robust to risk aversion under direct payment; its implementation cost is that exact-match payment probabilities can be low away from concentrated reports.

Second, use quadratic-distance frequency scoring when the subject is expected to place most of the probability on one or a few outcomes. It is the most informative rule in that regime—for the average-coordinate, worst-coordinate, and mean targets alike—and it offers a transparent projection rule, a linear identified set, closed-form coordinate bounds, and linear-program bounds for any linear functional.

Third, use Manhattan distance when it cannot be anticipated in advance whether the subject's beliefs will be concentrated on a few outcomes or spread across many. Its identified set is exact and its coordinate bounds are sharp—semi-analytical, reducing to a one-dimensional threshold computation rather than a closed form. Manhattan rarely gives the single narrowest interval, but the design comparison shows its regret is nearly constant across belief concentrations: it is never far from the best rule, and so minimizes worst-case regret for a researcher who does not know the regime. Its inferential interpretation is also median-oriented, which suits settings where a median-type count prediction is substantively appropriate.

5.1 Risk Aversion and Implementation

The body of the paper assumes risk neutrality. Whether that assumption can be relaxed depends on how the scoring rule is implemented as a payment. The distinction is between paying the score directly as money and using the score to determine the probability of a fixed prize.

Direct monetary scores. If the subject is paid the score itself, the subject solves

$$\arg\max_{r \in \Omega} \mathbb{E}_p[u(S(r,\omega))].$$

For the discrete-metric rule this does not change incentives. The payment is a fixed prize B if $r = \omega$ and zero otherwise, so

$$\mathbb{E}_p[u] = u(0) + \Pr_p(\omega = r)\{u(B) - u(0)\},$$

and any strictly increasing u preserves the same optimal reports as maximizing the exact-match probability. Direct monetary distance scores do not have this robustness: quadratic and Manhattan distance generate several possible payment levels, so nonlinear utility can change the ranking of reports. This is the standard risk-aversion problem for monetary scoring rules (Armantier and Treich, 2013; Schlag et al., 2015).

Probability implementation. The distance rules can instead be implemented as a binary lottery. Let $q(r,\omega) \in [0,1]$ be the score normalized into a probability, and pay a fixed prize B with probability $q(r,\omega)$ and zero otherwise. If the prize and outside payment are fixed, utility depends only on final money, and the randomization is objective and independent, then

$$\mathbb{E}_p[q(r,\omega)u(B) + (1 - q(r,\omega))u(0)] = u(0) + (u(B) - u(0))\,\mathbb{E}_p[q(r,\omega)],$$

so maximizing expected utility is equivalent to maximizing the expected normalized score, regardless of the curvature of u. A positive affine normalization of the bounded score into $[0,1]$ preserves the expected-score ranking of reports, so the body's risk-neutral analysis carries over unchanged. This is the binary-lottery, or binarized-scoring-rule, technique (Roth and Malouf, 1979; Berg et al., 1986; Karni, 2009; Hossain and Okui, 2013); the contribution here is to apply it, not to introduce it.

Two qualifications matter. First, the robustness is to risk aversion within expected utility. Subjects who depart from expected utility—for instance through probability weighting—can report non-truthfully even under the binary-lottery implementation, and the paper does not address such departures. Second, the binary-lottery implementation asks subjects to reason about a compound lottery on top of the count report, which partially offsets the cognitive simplicity that motivates frequency reports in the first place.

The implementation choice also interacts with a cost noted in the design comparison. The discrete-metric prize is paid only on an exact match, and the exact-match probability falls sharply as n, k, and belief balance increase, so at larger designs the rule becomes a lottery with a vanishing chance of any payment. The discrete-metric rule is thus robust to risk aversion even under direct payment but can become nearly unwinnable; the distance rules avoid that under the binary-lottery implementation, whose payment probability is the expected normalized score, at the cost of relying on expected utility.

5.2 Other Count-Loss Rules and the Limits of the Approach

The three rules compared above do not exhaust the family of count losses, and the partial-identification approach is not a general theory of discrete scoring rules. Two further rules—Hamming and Chebyshev distance—are worth stating precisely, because they mark where the approach stops yielding tractable sharp bounds.

Hamming distance. The Hamming rule $S_H(r,\omega) = a - b\sum_i \mathbf{1}\{r_i \neq \omega_i\}$ penalizes the number of categories whose reported count differs from the realized count. Its expected loss separates into binomial marginal exact-coordinate probabilities, $\mathbb{E}_p[\sum_i \mathbf{1}\{r_i \neq \omega_i\}] = \sum_i\{1 - \Pr(\mathrm{Bin}(n,p_i) = r_i)\}$, so an optimal report maximizes a separable objective and the identified set $P_H(r)$ is exactly the system of expected-loss inequalities over all feasible reports. For $k = 2$ the rule coincides with discrete-metric scoring and inherits its closed-form bounds. For $k > 2$ it does not. Each coordinate term $\Pr(\mathrm{Bin}(n,p_i) = r_i)$ is unimodal but not discrete-concave in the reported count, so a single-unit-transfer argument is necessary but not sufficient for global optimality: a multi-count transfer can move the report from one binomial mode to a higher one. This failure is not confined to the boundary of the simplex. At $n = 3$, $k = 5$, and uniform beliefs $p = (1/5,\ldots,1/5)$, for instance, the report $(3,0,0,0,0)$ admits no improving single transfer, yet the report $(0,0,1,1,1)$ yields a strictly higher expected score. The identified set $P_H(r)$ is therefore a non-convex semialgebraic set rather than a polytope. A closed-form modal-box inner bound and a single-unit-transfer outer bound sandwich it, but the sharp coordinate bounds lie strictly between these and are available only by numerical optimization over the non-convex set—which is computationally intractable at the category counts of the design comparison. Hamming distance is for this reason characterized here rather than included in the comparison.

Chebyshev distance. The Chebyshev rule $S_\infty(r,\omega) = a - b\max_i|r_i - \omega_i|$ penalizes the largest single-category count error. Unlike the other count losses, its expected loss does not separate across coordinates: the maximum couples the category errors within every realization. For $k = 2$ the rule reduces to Manhattan scoring—the two losses differ only by a positive factor—and inherits the binomial-median interval. For $k > 2$ no clean optimal-report characterization is available; the identified set is again only the system of expected-loss inequalities over all reports, with no transfer-polytope or threshold reduction. Chebyshev distance is computable by enumeration for small n and k but, like Hamming, does not deliver the sharp, scalable bounds that make the three headline rules comparable.

These two rules are not pathological—their identified sets are well defined and finite to check—but they show that the closed-form and semi-analytical tractability of the headline rules is a property of those rules, not a generic feature of count-loss scoring.

5.3 An Open Empirical Question

The case for frequency reports rests partly on a behavioural premise this paper does not test: that subjects reason more readily in counts than in abstract probabilities, and find a distance-based payment more transparent than a probability score. That premise is taken here on the strength of the cited literature. The partial-identification framework itself, however, suggests how it could be examined. If beliefs are induced by the experimenter—through draws from an urn of known composition, or a signal structure with a computable posterior—the true belief p is known, so an elicited report r can be checked against the identified set $P_S(r)$ it generates: under optimal reporting $P_S(r)$ contains p, and the empirical coverage rate, with the direction of any systematic miss, measures how far subjects depart from optimal reporting. Eliciting the same induced beliefs under a frequency-report rule and under a standard probability scoring rule isolates the effect of the report format; crossing that with direct versus binary-lottery payment isolates the effect of the implementation, which the risk-aversion analysis above predicts to matter only for subjects who are not risk neutral. Response times, incentivized comprehension questions, and the rate of weakly dominated reports would complement the coverage measure. Such a design would test, rather than presume, the behavioural premise on which the case for frequency reports partly rests—a premise separate from the present contribution, which is that different frequency-report scoring rules induce different identified sets, and those sets determine the informational efficiency of the mechanism.

A Proofs

A.1 Discrete-Metric Frequency Scoring

Proof of Proposition 2. For any report r,

$$\mathbb{E}_p[S_0(r, \omega)] = a - b \Pr_p(\omega \neq r) = (a - b) + b \Pr_p(\omega = r),$$

so, since $b > 0$, expected-score maximization is equivalent to maximizing the multinomial probability mass $\Pr_p(\omega = r)$.

It remains to characterize the mode region. Compare r with $r + e_i - e_j$, which is feasible when $r_j > 0$. If $p_j = 0$ and $r_j > 0$, then r cannot be a mode unless all probability is zero, which is impossible. For $p_j > 0$,

$$\frac{\Pr_p(r + e_i - e_j)}{\Pr_p(r)} = \frac{r_j}{r_i + 1} \frac{p_i}{p_j}.$$

Thus a necessary condition for r to be a mode is

$$r_j p_i \le (r_i + 1) p_j \quad \forall i, j \text{ with } r_j > 0.$$

Conversely, if these inequalities hold, any feasible s can be reached from r by a sequence of one-count transfers from coordinates where the current vector exceeds s to coordinates where it is below s. At each step the same ratio is at most one, because source counts weakly decrease and destination counts weakly increase along the path. Therefore no feasible s has larger multinomial mass than r.

The coordinate upper bound follows by summing

$$r_j p_i \le (r_i + 1) p_j$$

over $j \neq i$:

$$(n - r_i) p_i \le (r_i + 1)(1 - p_i),$$

so $p_i \le (r_i + 1)/(n + 1)$. The lower bound follows by summing

$$r_i p_j \le (r_j + 1) p_i$$

over $j \neq i$:

$$r_i (1 - p_i) \le (n - r_i + k - 1) p_i,$$

so $p_i \ge r_i/(n + k - 1)$.

The bounds are sharp. The upper endpoint is attained by

$$p_i = \frac{r_i + 1}{n + 1}, \quad p_j = \frac{r_j}{n + 1} \quad (j \neq i).$$

The lower endpoint, when $r_i > 0$, is attained by

$$p_i = \frac{r_i}{n + k - 1}, \quad p_j = \frac{r_j + 1}{n + k - 1} \quad (j \neq i).$$

When $r_i = 0$, the lower endpoint 0 is attained by setting $p_i = 0$ and assigning probabilities proportional to r_j on the reported support. When $r_i = n$, the upper endpoint 1 is attained by $p_i = 1$. □

A.2 Manhattan-Distance Rule

The Manhattan characterization used in Section 3.2 follows from a standard exchange argument. Linearity of expectation gives

$$\mathbb{E}_p \left[\sum_{i=1}^{k} |r_i - \omega_i| \right] = \sum_{i=1}^{k} \mathbb{E}_p |r_i - \omega_i|.$$

The i-th term depends only on the binomial marginal $W_i \sim \mathrm{Bin}(n, p_i)$. For

$$\ell_i(t; p_i) = \mathbb{E}|t - W_i|,$$

the marginal cost of increasing the reported count from t to $t + 1$ is

$$\ell_i(t + 1; p_i) - \ell_i(t; p_i) = 2F_i(t; n, p_i) - 1.$$

Moving one reported count from j to i changes expected Manhattan loss by

$$\{2F_i(r_i;n,p_i)-1\}-\{2F_j(r_j-1;n,p_j)-1\}.$$

Hence no feasible one-count transfer improves the report if and only if

$$F_i(r_i;n,p_i)\geq F_j(r_j-1;n,p_j)\quad \forall i,j \text{ with } r_i<n,\ r_j>0.$$

This local no-transfer condition is also sufficient for global optimality. Each coordinate loss ℓ_i is discrete-convex—its increments $2F_i(t;n,p_i)-1$ are weakly increasing in t—and the objective $\sum_i \ell_i(r_i;p_i)$ is minimized over the integer simplex $\{r:\sum_i r_i=n\}$. Suppose r satisfies the no-transfer condition and let s be any feasible report. The difference $s-r$ sums to zero, so s is reached from r by a sequence of single-count transfers; choose the sequence monotone, at each step moving a unit from a coordinate currently above its target s to one currently below. A coordinate below its target only ever receives counts and so remains below it, and a coordinate above its target only ever sends counts and remains above it; hence each coordinate's count moves monotonically between r_i and s_i, and every intermediate report is feasible. At a step that moves a unit from j to i, the receiving count c_i satisfies $c_i\geq r_i$ and the sending count c_j satisfies $c_j\leq r_j$; by discrete-convexity the step's cost $\{\ell_i(c_i+1)-\ell_i(c_i)\}+\{\ell_j(c_j-1)-\ell_j(c_j)\}$ is at least the cost of the corresponding transfer evaluated at r, which is nonnegative because r satisfies the no-transfer condition. Summing over the path gives $\sum_i \ell_i(s_i)\geq\sum_i \ell_i(r_i)$, so r is globally optimal. Hence $P_M(r)$ is the exact identified set.

The threshold representation follows by taking c between

$$\max_{j:r_j>0} F_j(r_j-1;n,p_j)\quad\text{and}\quad \min_{i:r_i<n} F_i(r_i;n,p_i).$$

Inverting the resulting binomial-CDF inequalities coordinate by coordinate gives the threshold-indexed box-simplex description used for Manhattan bounds.

A.3 Quadratic-Distance Rule

Proof of Proposition 1. Maximizing S_2 is equivalent to minimizing $\mathbb{E}_p\|r-\omega\|_2^2$. For fixed r,

$$\mathbb{E}_p\|r-\omega\|_2^2=\|r-\mathbb{E}_p[\omega]\|_2^2+\sum_{i=1}^k \mathrm{Var}_p(\omega_i).$$

The variance term is independent of r, and $\mathbb{E}_p[\omega]=np$. Hence optimal reports are exactly the feasible integer projections of np.

The full inverse projection condition is

$$\|r-np\|_2^2\leq\|s-np\|_2^2\quad\forall s\in\Omega,$$

or equivalently

$$2np\cdot(s-r)\leq\|s\|_2^2-\|r\|_2^2.$$

Taking $s=r+e_i-e_j$, feasible when $r_j>0$, gives

$$n(p_i-p_j)\leq r_i-r_j+1.$$

Conversely, suppose all these one-count inequalities hold. For any $s \in \Omega$, write $d = s - r$. Pair each positive unit of d_i with a negative unit of d_j. Each coordinate j with $d_j < 0$ has $s_j < r_j$ and hence $r_j > 0$, so the one-count inequality $n(p_i - p_j) \le r_i - r_j + 1$ is available for every paired transfer. Summing the one-count inequalities over the paired transfers gives

$$np \cdot d \le r \cdot d + M,$$

where $M = \sum_{i:d_i>0} d_i$ is the number of transferred counts. Because d has integer components, $|d_i| \ge 1$ on its support, so

$$\sum_i d_i^2 \ge \sum_i |d_i| = 2M.$$

Doubling the displayed inequality and substituting $2M \le \sum_i d_i^2 = \|d\|_2^2$ gives $2np \cdot d \le 2r \cdot d + \|d\|_2^2$. Since $d = s - r$,

$$2np \cdot (s - r) \le 2r \cdot (s - r) + \|s - r\|_2^2 = \|s\|_2^2 - \|r\|_2^2,$$

which is the full projection condition.

It remains to derive the coordinate bounds. Let $m = m(r) = |\{j : r_j > 0\}|$. If $r_i > 0$, summing

$$n(p_u - p_i) \le r_u - r_i + 1$$

over all $u \ne i$ gives

$$1 - p_i \le (k-1)p_i + \frac{n - kr_i + k - 1}{n},$$

so

$$p_i \ge \frac{r_i - 1}{n} + \frac{1}{nk}.$$

For the upper bound, sum

$$n(p_i - p_v) \le r_i - r_v + 1$$

over positive-report coordinates $v \ne i$, and use nonnegativity of the zero-report coordinates:

$$1 - p_i \ge (m-1)p_i + \frac{n - mr_i - m + 1}{n}.$$

This gives

$$p_i \le \frac{r_i + 1}{n} - \frac{1}{nm}.$$

If $r_i = 0$, the lower bound is 0. For the upper bound, sum

$$n(p_i - p_v) \le 1 - r_v$$

over all m positive-report coordinates v:

$$1 - p_i \ge mp_i + \frac{n - m}{n},$$

so

$$p_i \le \frac{m}{n(m+1)}.$$

The bounds are sharp, with an explicit attaining belief for each endpoint. For the lower bound with $r_i > 0$, take $p_i = (r_i - 1)/n + 1/(nk)$ and $p_u = r_u/n + 1/(nk)$ for $u \ne i$. This is a probability vector—it sums to one and every coordinate is at least $1/(nk)$—and $n(p_a - p_b) = r_a - r_b \le r_a - r_b + 1$ for $a, b \ne i$ while $n(p_a - p_i) = r_a - r_i + 1$, so it lies in $P_Q(r)$ and the constraints with $b = i$ bind, attaining the lower endpoint. For the upper bound with $r_i > 0$, take $p_i = (r_i + 1)/n - 1/(nm)$, take $p_v = r_v/n - 1/(nm)$ for the positive-report coordinates $v \ne i$, and set $p_v = 0$ for the zero-report coordinates. This sums to one and is nonnegative, lies in $P_Q(r)$, and makes the constraints $n(p_i - p_v) \le r_i - r_v + 1$ bind for every positive-report $v \ne i$, attaining the upper endpoint. For the upper bound with $r_i = 0$, take $p_i = m/(n(m+1))$, take $p_v = \{(m+1)r_v - 1\}/(n(m+1))$ for the positive-report coordinates v, and set $p_w = 0$ for the other zero-report coordinates. This sums to one and is nonnegative, lies in $P_Q(r)$, and makes the constraints $n(p_i - p_v) \le 1 - r_v$ bind, attaining the upper endpoint. Thus the displayed bounds are sharp. □

B Design Comparison Details

The design comparison is generated by the script `scripts/design_efficiency.py`. The final grid uses $n \in \{5, 10, 20, 50\}$, $k \in \{2, 3, 5, 10\}$, $\alpha \in \{0.1, 0.3, 1, 3, 10\}$, and 5,000 draws from the symmetric Dirichlet distribution for each cell. For each draw, and for each of the three rules in the comparison—discrete-metric, quadratic-distance, and Manhattan—the script computes an optimal report and then bounds for the identified set generated by that report. Hamming and Chebyshev distance are excluded; Section 5 explains why their identified sets do not yield tractable sharp bounds.

Discrete-metric coordinate bounds are closed form and mean bounds are linear programs over the transfer region. Quadratic-distance coordinate bounds are closed form and mean bounds are linear programs over the one-count transfer polytope. Manhattan coordinate bounds are sharp: the script solves the one-dimensional threshold optimization implied by Appendix A.2 with a monotone root-find. Manhattan mean bounds use the same threshold representation but are computed over a threshold grid with tolerance 10^{-4}.

The generated files are CSV tables under `outputs/design_exercise/`. These outputs are diagnostics for experimental design. They are not simulations of subject behavior and do not replace the mathematical characterization of $R_S(p)$ and $P_S(r)$.

Table 3 substantiates the claim in Section 4 that the number of categories k and the number of trials n are second-order. It reports win shares marginally by k and by n, averaging over the other two design dimensions. Against the variation across belief concentration α in Table 1—where the leading rule's win share runs from near zero to one—the dependence on k and n is slight.

Robustness. Two implementation choices were checked and leave the comparison unchanged. First, the linear-functional results are robust to the outcome vector. The same run also evaluates the extreme-category outcome vector $x = (0, \dots, 0, 1)$ in place of the evenly spaced $x_i = (i-1)/(k-1)$; the ranking is unchanged, with win shares 0.56, 0.34, and 0.10 for discrete-metric, quadratic-distance, and Manhattan scoring, against 0.53, 0.38, and 0.10 for the evenly spaced vector. Second, optimal-report ties are a measure-zero event under the continuous Dirichlet draws: the final run recorded a tie rate of 0.0000 for all three rules, so the lexicographic convention used to select a report when $R_S(p)$ is not a singleton has no effect on the reported numbers.

	Average-coordinate			Linear-mean		
	Discrete	Quadratic	Manhattan	Discrete	Quadratic	Manhattan
By number of categories k						
2	0.72	0.25	0.03	0.72	0.25	0.03
3	0.59	0.37	0.04	0.34	0.51	0.15
5	0.58	0.35	0.07	0.49	0.42	0.09
10	0.59	0.31	0.10	0.56	0.33	0.11
By number of trials n						
5	0.60	0.34	0.06	0.57	0.33	0.10
10	0.61	0.32	0.07	0.57	0.33	0.11
20	0.62	0.31	0.06	0.53	0.38	0.09
50	0.65	0.30	0.05	0.44	0.47	0.09

Table 3: Win share marginally by number of categories k and number of trials n, averaging over the other two design dimensions. Relative to belief concentration (Table 1), the dependence is second-order: the coordinate target is nearly flat in both, and the only visible movements are the binary case $k = 2$ and a mild drift toward quadratic-distance scoring at large n for the mean target.

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